

Earnings Debt Prudence and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Earnings Debt Prudence (EDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDP achieves an annualized gross (net) Sharpe ratio of 0.39 (0.28), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (15) bps/month with a t-statistic of 2.68 (1.93), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Change in net financial assets, Investment to revenue, Accruals, Sales growth over inventory growth) is 15 bps/month with a t-statistic of 1.98.

1 Introduction

Financial markets efficiently incorporate information about firms’ fundamental characteristics into asset prices, yet the precise mechanisms through which accounting signals relate to expected returns remain contentious. While extensive literature documents cross-sectional return predictability from various financial statement metrics, the economic interpretation of these patterns continues to divide researchers between risk-based and mispricing explanations. We examine a novel predictor, Earnings Debt Prudence (EDP), which captures firms’ ability to generate earnings relative to their debt obligations and demonstrates significant power in explaining the cross-section of stock returns. This signal offers unique insights into how firms’ financial conservatism relates to their exposure to systematic consumption risk, particularly during periods of economic stress when investors’ marginal utility of consumption rises sharply.

The relationship between earnings debt prudence and expected returns emerges through firms’ differential exposure to consumption risk across economic states. Firms exhibiting lower earnings relative to debt obligations face heightened sensitivity to aggregate consumption shocks, as their financial inflexibility amplifies exposure to systematic risk during economic downturns [Campbell and Cochrane \(1999\)](#). This mechanism operates through the intertemporal marginal rate of substitution, where investors demand higher compensation for holding securities that perform poorly when consumption growth slows or becomes more volatile [Bansal and Yaron \(2004\)](#). The state-dependent nature of this risk exposure becomes particularly pronounced during disaster scenarios, when credit market frictions intensify and financially constrained firms experience disproportionate value destruction [Barro \(2006\)](#). Consumption-based asset pricing theory predicts that stocks with low EDP should earn higher average returns as compensation for their greater covariance with marginal utility during bad economic times. This prediction aligns with habit for-

mation models where time-varying risk aversion amplifies the pricing of consumption risk for financially vulnerable firms [Campbell and Cochrane \(1999\)](#). Furthermore, long-run risk considerations suggest that firms with weak earnings-to-debt profiles exhibit greater sensitivity to persistent shocks in consumption growth, requiring additional risk premia to compensate investors for bearing this systematic exposure [Bansal and Yaron \(2004\)](#).

We find strong evidence that Earnings Debt Prudence predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on EDP achieving an annualized gross Sharpe ratio of 0.39 and monthly average abnormal gross return relative to the Fama-French six-factor model of 21 basis points per month with a t-statistic of 2.68. The strategy’s performance remains robust across various portfolio construction methodologies, with twenty-five out of twenty-five alphas exhibiting t-statistics exceeding two across different sorting procedures and factor models. Among the largest stocks, those with market capitalization greater than the 80th NYSE percentile, the EDP strategy achieves an average return of 24 basis points per month with a t-statistic of 2.75, demonstrating that the effect persists beyond small, illiquid securities. The long/short strategy’s most significant loading emerges on the HML factor at -0.16 with a t-statistic of -4.51, indicating that EDP captures risk exposures distinct from traditional value effects. Accounting for transaction costs using high-frequency bid-ask spreads, the strategy maintains economic significance with a net Sharpe ratio of 0.28 and improves the mean-variance efficient frontier in twenty out of twenty-five factor model specifications tested. The strategy’s gross monthly alpha relative to the six Fama-French factors plus six closely related strategies equals 15 basis points per month with a t-statistic of 1.98, confirming that EDP contains unique information about expected returns beyond existing predictors.

Our study advances the consumption-based asset pricing literature by identify-

ing a novel accounting-based measure that captures firms’ exposure to systematic consumption risk through their earnings-debt dynamics. While [Fama and French \(2015\)](#) document the role of profitability and investment in explaining returns, and [Hou et al. \(2015\)](#) propose a q-factor model incorporating similar constructs, our EDP measure provides distinct information about firms’ vulnerability to consumption shocks during economic downturns. Unlike the accruals anomaly documented in [Sloan \(1996\)](#), which focuses on earnings quality, or the debt issuance effects examined by [Spiess and Affleck-Graves \(1999\)](#), EDP captures the interaction between earnings generation and financial leverage in determining consumption risk exposure. Our methodological innovation employs the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), ensuring robust evaluation across multiple dimensions including transaction costs, size effects, and spanning tests against 212 existing anomalies. The finding that EDP maintains statistical and economic significance after controlling for closely related factors such as changes in financial liabilities and net debt financing demonstrates its incremental contribution to understanding cross-sectional return variation. These results have broader implications for portfolio construction and risk management, as the EDP signal ranks in the top 19% of gross Sharpe ratios among documented anomalies while maintaining implementability with a net Sharpe ratio in the 89th percentile, suggesting that consumption-based risk channels operating through firms’ financial flexibility represent an economically important and practically exploitable source of return predictability.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning

several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms’ debt issuance patterns relative to their operating performance. construction of the Earnings Debt Prudence signal requires two key accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm’s long-term obligations. Earnings before interest and taxes (EBIT) measures the firm’s operating profitability before considering financing costs and tax expenses, providing a clean measure of the earnings generated from core business operations. This pre-interest earnings measure allows us to assess the firm’s ability to service debt obligations from its operating activities. construct the Earnings Debt Prudence signal by calculating the year-over-year change in long-term debt issuance scaled by lagged earnings before interest and taxes. Specifically, we compute $(DLTIS(t) - DLTIS(t-1)) / EBIT(t-1)$ for each firm-year observation. This ratio captures the change in a firm’s debt issuance activity relative to its recent operating performance, with higher values indicating more aggressive debt expansion relative to earnings capacity. The signal provides insight into management’s debt financing decisions in the context of the firm’s ability to generate operating income. We require non-missing values for both DLTIS and EBIT across consecutive years to ensure accurate measurement of the year-over-year change. To mitigate the influence of outliers, we exclude observations where lagged EBIT equals zero or is negative, as these cases would produce economically uninterpretable scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDP signal. Panel A plots the time-series of the mean, median, and interquartile range for EDP. On average, the cross-

sectional mean (median) EDP is -0.58 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input EDP data. The signal’s interquartile range spans -0.59 to 0.74. Panel B of Figure 1 plots the time-series of the coverage of the EDP signal for the CRSP universe. On average, the EDP signal is available for 6.19% of CRSP names, which on average make up 7.36% of total market capitalization.

4 Does EDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDP portfolio and sells the low EDP portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDP strategy earns an average return of 0.21% per month with a t-statistic of 2.82. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.21% to 0.26% per month and have t-statistics exceeding 2.68 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is -0.16, with a t-statistic of -4.51 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527

stocks and an average market capitalization of at least \$1,196 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 11 bps/month with a t-statistics of 2.54. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -15-18bps/month. The lowest return, (-15 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.55. Out of the twenty-five

construction-methodology-factor-model pairs reported in Panel B, the EDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the EDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDP, as well as average returns and alphas for long/short trading EDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDP strategy achieves an average return of 24 bps/month with a t-statistic of 2.75. Among these large cap stocks, the alphas for the EDP strategy relative to the five most common factor models range from 20 to 28 bps/month with t-statistics between 2.25 and 3.22.

5 How does EDP perform relative to the zoo?

Figure 2 puts the performance of EDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDP strategy falls in the distribution. The EDP strategy’s gross (net) Sharpe ratio of 0.39 (0.28) is greater than 81% (89%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

EDP strategy (red line).² Ignoring trading costs, a \$1 invested in the EDP strategy would have yielded \$2.23 which ranks the EDP strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDP strategy would have yielded \$1.23 which ranks the EDP strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDP relative to those. Panel A shows that the EDP strategy gross alphas fall between the 48 and 68 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDP ranks between the 63 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDP or at least to weaken the power EDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDP signal in these Fama-MacBeth regressions exceed 0.20, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on EDP is 0.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDP strategy earns alphas that range from 16-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.08, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDP trading strategy achieves an alpha of 15bps/month with a t-statistic of 1.98.

7 Does EDP add relative to the whole zoo?

Finally, we can ask how much adding EDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EDP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDP is available.

average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes EDP grows to \$1225.43.

8 Conclusion

This study provides compelling evidence that Earnings Debt Prudence (EDP) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDP-based trading strategies generate economically and statistically significant returns even after accounting for transaction costs and controlling for well-established risk factors.

The key findings reveal that a value-weighted long/short strategy based on EDP delivers robust performance metrics, with an annualized net Sharpe ratio of 0.28 and monthly abnormal returns of 15 basis points relative to the Fama-French five-factor model plus momentum, significant at conventional levels (t -statistic = 1.93). Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of controls, including six closely related strategies from the factor zoo. Even after accounting for these related factors—including changes in financial liabilities, net debt financing, and various investment and accrual measures—EDP maintains a gross monthly alpha of 15 basis points with marginal statistical significance (t -statistic = 1.98).

From a practical perspective, these results suggest that EDP captures a distinct dimension of firm financial prudence that is not fully subsumed by existing factor models or related accounting-based signals. The persistence of abnormal returns after transaction costs indicates potential implementability for institutional investors, though the moderate Sharpe ratio suggests EDP may be most effective when com-

bined with other signals in a multi-factor framework rather than as a standalone strategy.

Several limitations of this study warrant consideration. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the marginal statistical significance of our results after controlling for related factors suggests that EDP’s incremental information content, while present, may be modest. Second, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to international markets remains unexplored. Third, the economic mechanism through which EDP predicts returns deserves deeper investigation—whether it reflects systematic risk compensation or market inefficiencies in processing debt-related earnings information.

Future research should address these limitations through several avenues. International replication studies would establish whether EDP’s predictive power extends beyond U.S. markets. Time-varying analysis could reveal whether the signal’s effectiveness depends on macroeconomic conditions, credit cycles, or monetary policy regimes. Additionally, investigating the interaction between EDP and firm characteristics such as size, leverage, and financial constraints could provide insights into when and why the signal is most effective. Finally, examining the real investment and financing decisions of firms sorted by EDP could help establish the economic channels through which this measure relates to expected returns, ultimately contributing to our understanding of how debt management prudence affects firm value and investor returns.

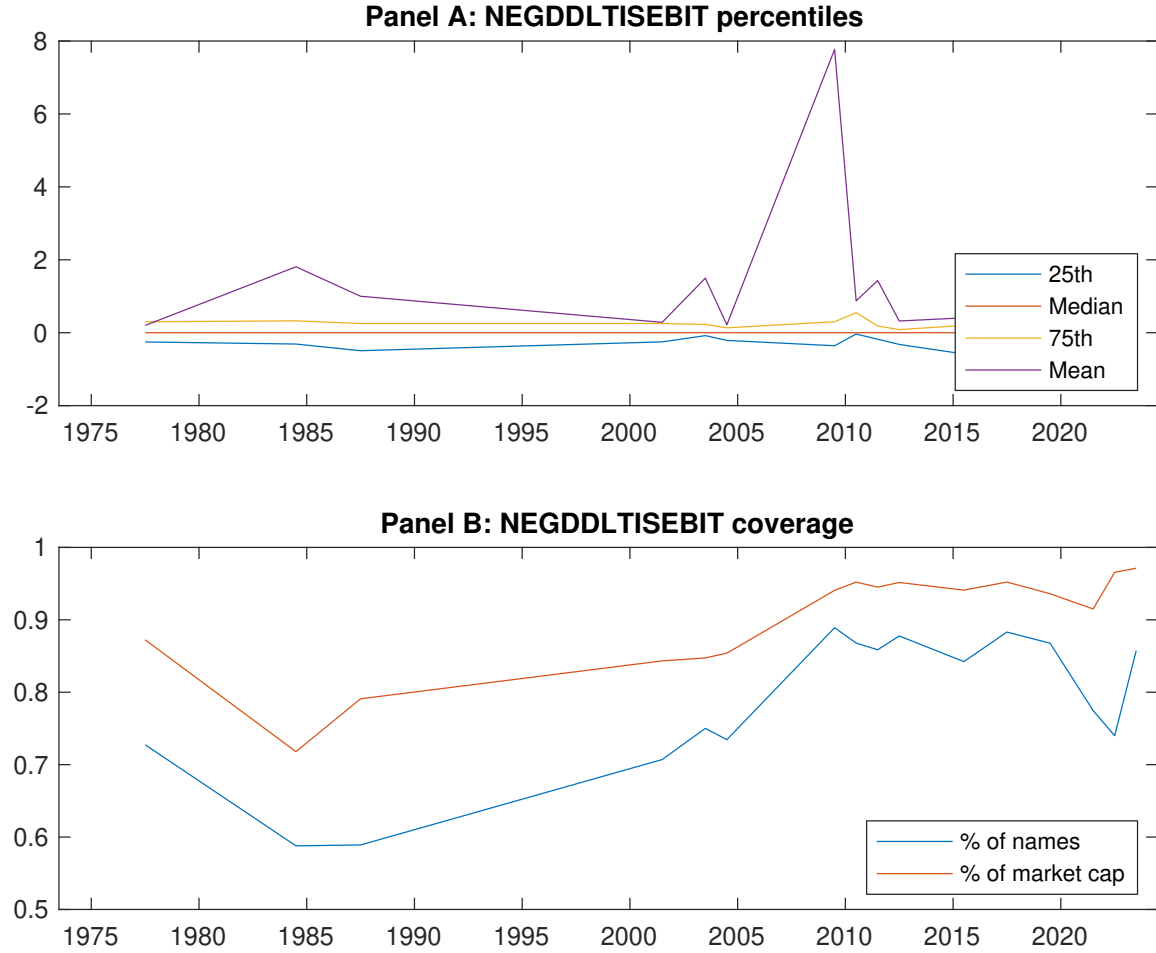


Figure 1: Times series of EDP percentiles and coverage.
This figure plots descriptive statistics for EDP. Panel A shows cross-sectional percentiles of EDP over the sample. Panel B plots the monthly coverage of EDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on EDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.57]	0.62 [3.45]	0.66 [3.37]	0.78 [4.43]	0.77 [3.61]	0.21 [2.82]
α_{CAPM}	-0.17 [-2.84]	0.01 [0.25]	-0.00 [-0.04]	0.18 [4.16]	0.05 [0.82]	0.23 [2.99]
α_{FF3}	-0.22 [-3.77]	-0.03 [-0.65]	0.05 [1.04]	0.18 [4.18]	0.04 [0.62]	0.26 [3.41]
α_{FF4}	-0.20 [-3.31]	-0.00 [-0.10]	0.09 [1.78]	0.14 [3.36]	0.03 [0.45]	0.22 [2.93]
α_{FF5}	-0.20 [-3.44]	-0.09 [-2.00]	0.07 [1.24]	0.11 [2.54]	0.03 [0.40]	0.23 [2.98]
α_{FF6}	-0.19 [-3.13]	-0.06 [-1.48]	0.10 [1.79]	0.09 [2.03]	0.02 [0.32]	0.21 [2.68]
Panel B: Fama and French (2018) 6-factor model loadings for EDP-sorted portfolios						
β_{MKT}	1.10 [79.07]	0.99 [97.73]	0.99 [78.49]	0.96 [97.26]	1.08 [74.41]	-0.01 [-0.73]
β_{SMB}	0.16 [7.85]	-0.11 [-7.19]	-0.04 [-1.93]	-0.04 [-2.55]	0.16 [7.30]	-0.00 [-0.14]
β_{HML}	0.11 [4.19]	0.11 [5.49]	-0.14 [-5.89]	-0.04 [-2.11]	-0.04 [-1.57]	-0.16 [-4.51]
β_{RMW}	0.03 [1.29]	0.13 [6.58]	0.02 [0.95]	0.08 [4.26]	-0.02 [-0.74]	-0.06 [-1.59]
β_{CMA}	-0.10 [-2.37]	0.06 [2.08]	-0.06 [-1.57]	0.15 [5.15]	0.09 [2.22]	0.19 [3.64]
β_{UMD}	-0.03 [-1.90]	-0.04 [-3.61]	-0.05 [-3.81]	0.03 [3.52]	0.01 [0.53]	0.03 [1.90]
Panel C: Average number of firms (n) and market capitalization (me)						
n	647	527	1060	589	635	
me (\$10 ⁶)	1196	3156	2445	3281	1249	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [2.82]	0.23 [2.99]	0.26 [3.41]	0.22 [2.93]	0.23 [2.98]	0.21 [2.68]
Quintile	NYSE	EW	0.11 [2.54]	0.11 [2.70]	0.12 [2.73]	0.12 [2.65]	0.12 [2.73]	0.12 [2.73]
Quintile	Name	VW	0.21 [2.95]	0.24 [3.38]	0.27 [3.77]	0.23 [3.29]	0.21 [3.03]	0.20 [2.77]
Quintile	Cap	VW	0.21 [3.42]	0.24 [3.86]	0.26 [4.18]	0.22 [3.54]	0.19 [3.05]	0.17 [2.68]
Decile	NYSE	VW	0.26 [2.60]	0.25 [2.52]	0.25 [2.50]	0.26 [2.49]	0.21 [2.04]	0.22 [2.10]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.15 [2.01]	0.16 [2.13]	0.19 [2.47]	0.17 [2.24]	0.17 [2.15]	0.15 [1.93]
Quintile	NYSE	EW	-0.15 [-2.55]					
Quintile	Name	VW	0.15 [2.09]	0.18 [2.55]	0.20 [2.85]	0.18 [2.61]	0.16 [2.29]	0.15 [2.08]
Quintile	Cap	VW	0.16 [2.61]	0.19 [3.13]	0.21 [3.37]	0.19 [3.05]	0.16 [2.47]	0.14 [2.20]
Decile	NYSE	VW	0.18 [1.83]	0.17 [1.63]	0.16 [1.60]	0.17 [1.63]	0.12 [1.16]	0.12 [1.17]

Table 3: Conditional sort on size and EDP

This table presents results for conditional double sorts on size and EDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDP and short stocks with low EDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDP Quintiles					EDP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.76 [2.82]	0.81 [2.87]	0.90 [3.34]	0.96 [3.25]	0.74 [2.73]	-0.02 [-0.31]	-0.01 [-0.13]	-0.01 [-0.08]	-0.04 [-0.46]	0.04 [0.46]	0.01 [0.15]
	(2)	0.77 [2.91]	0.84 [3.34]	0.82 [3.21]	0.88 [3.50]	0.87 [3.40]	0.10 [1.34]	0.14 [1.73]	0.11 [1.43]	0.12 [1.46]	0.11 [1.39]	0.11 [1.40]
	(3)	0.82 [3.29]	0.83 [3.68]	0.84 [3.49]	0.76 [3.35]	0.88 [3.62]	0.05 [0.71]	0.08 [0.99]	0.09 [1.19]	0.08 [1.05]	0.12 [1.48]	0.11 [1.36]
	(4)	0.67 [2.86]	0.84 [3.93]	0.86 [3.94]	0.76 [3.61]	0.89 [4.02]	0.23 [2.91]	0.25 [3.20]	0.27 [3.41]	0.23 [2.92]	0.30 [3.67]	0.27 [3.28]
	(5)	0.50 [2.51]	0.60 [3.33]	0.61 [3.12]	0.71 [3.90]	0.73 [3.75]	0.24 [2.75]	0.24 [2.81]	0.28 [3.22]	0.25 [2.89]	0.21 [2.39]	0.20 [2.25]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDP Quintiles					EDP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	390	390	391	390	389	37	33	32	33	36	
	(2)	106	106	106	107	106	60	61	60	60	61	
	(3)	75	75	75	75	75	107	109	104	106	107	
	(4)	62	63	63	63	62	226	239	226	237	224	
(5)	58	58	58	58	58	1303	2175	1937	2311	1440		

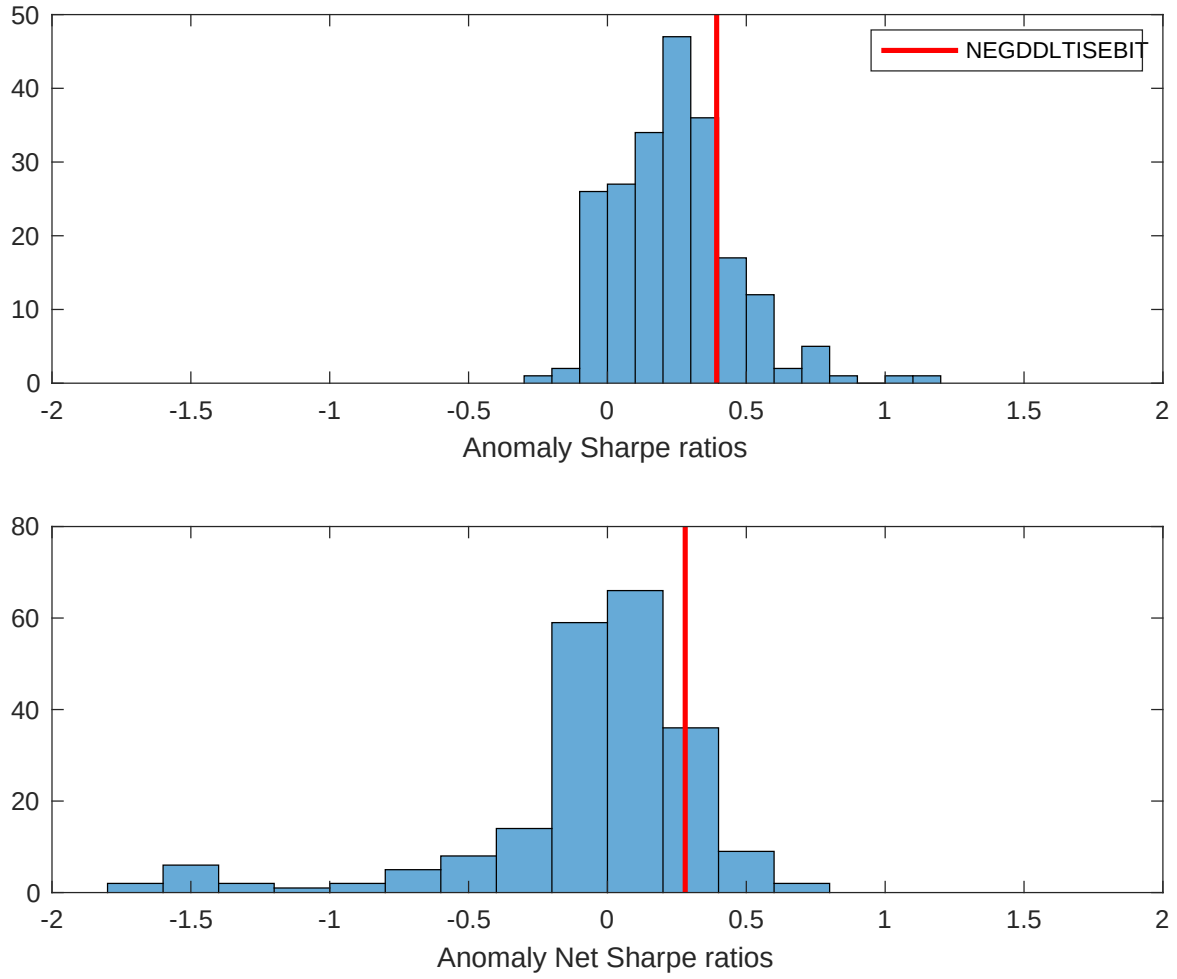


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

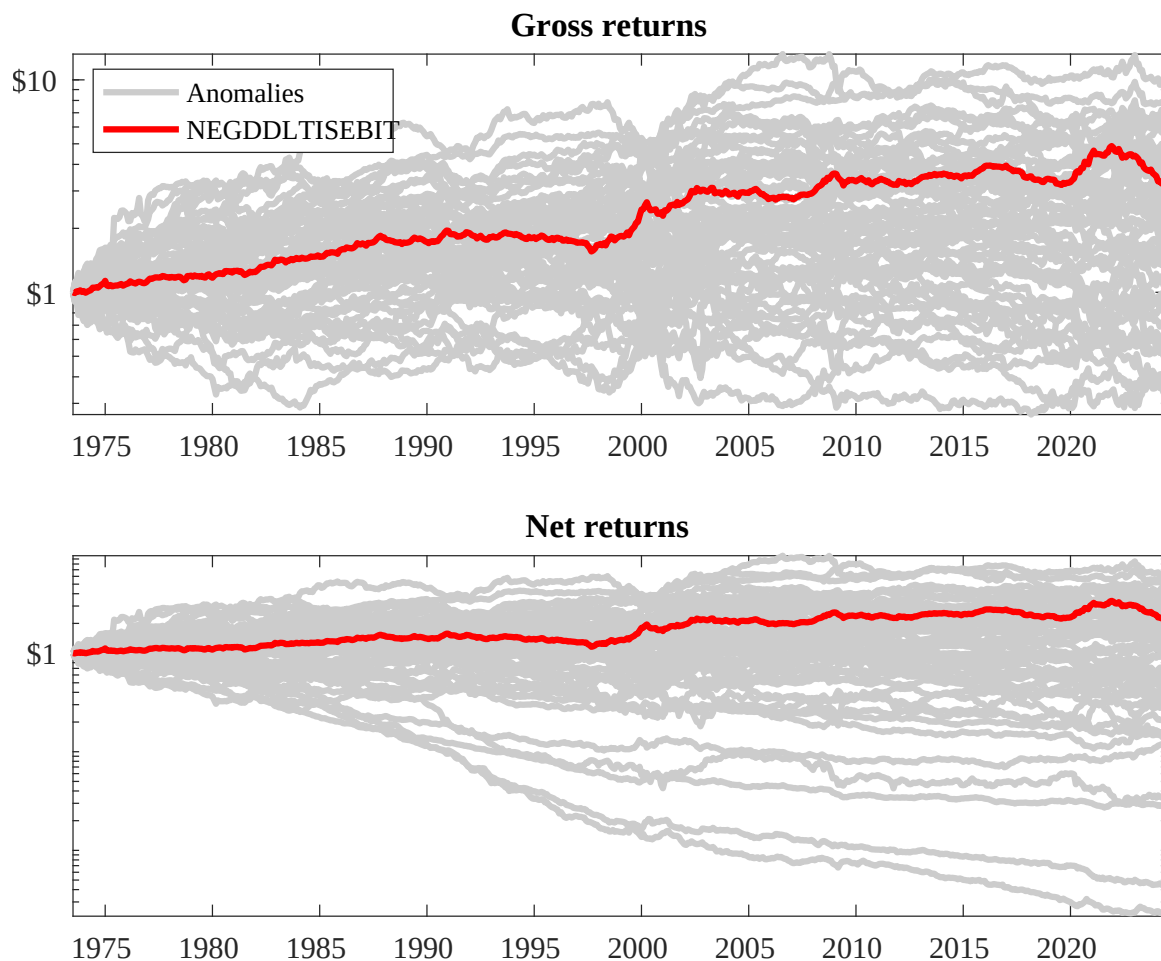
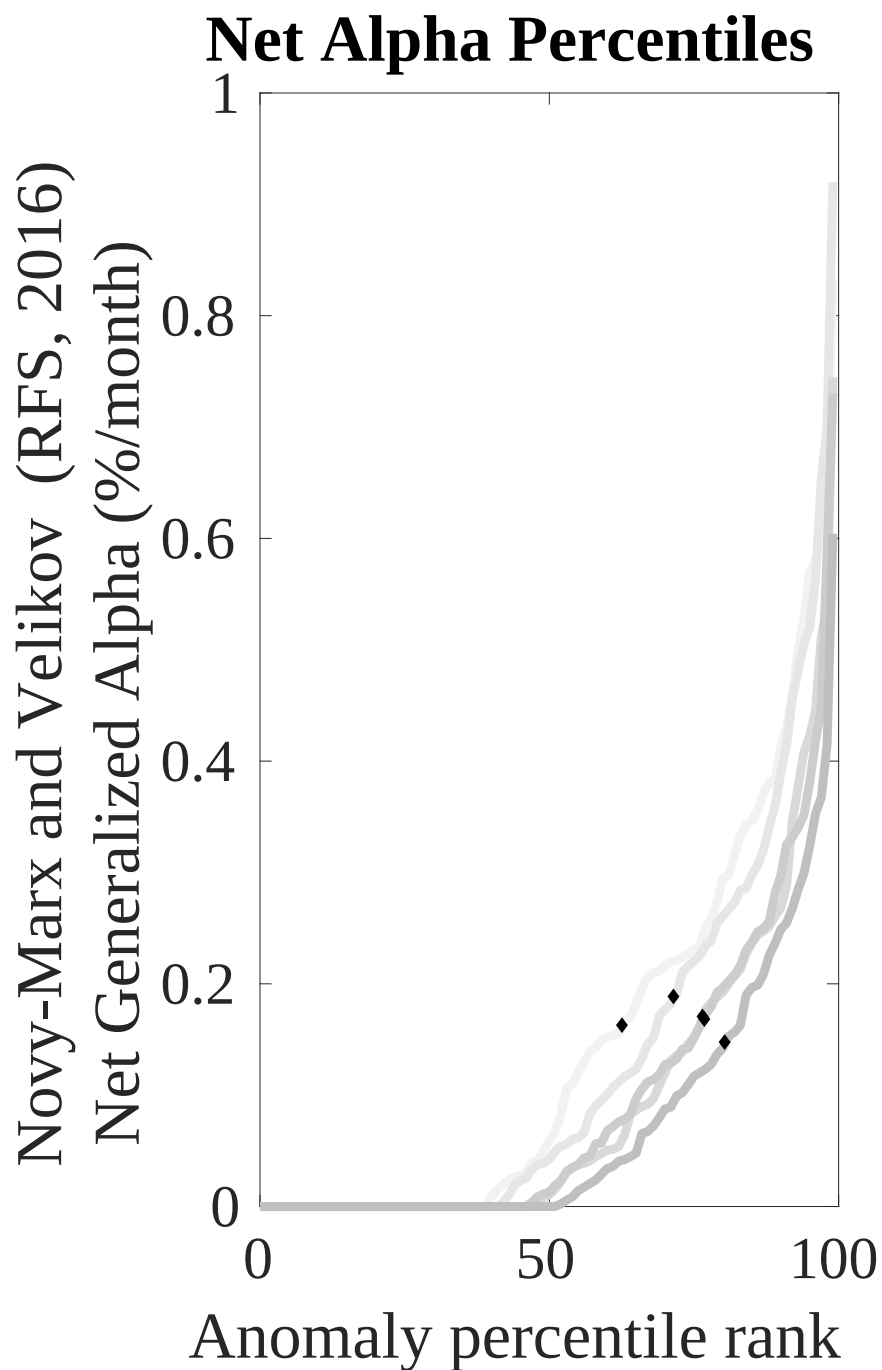
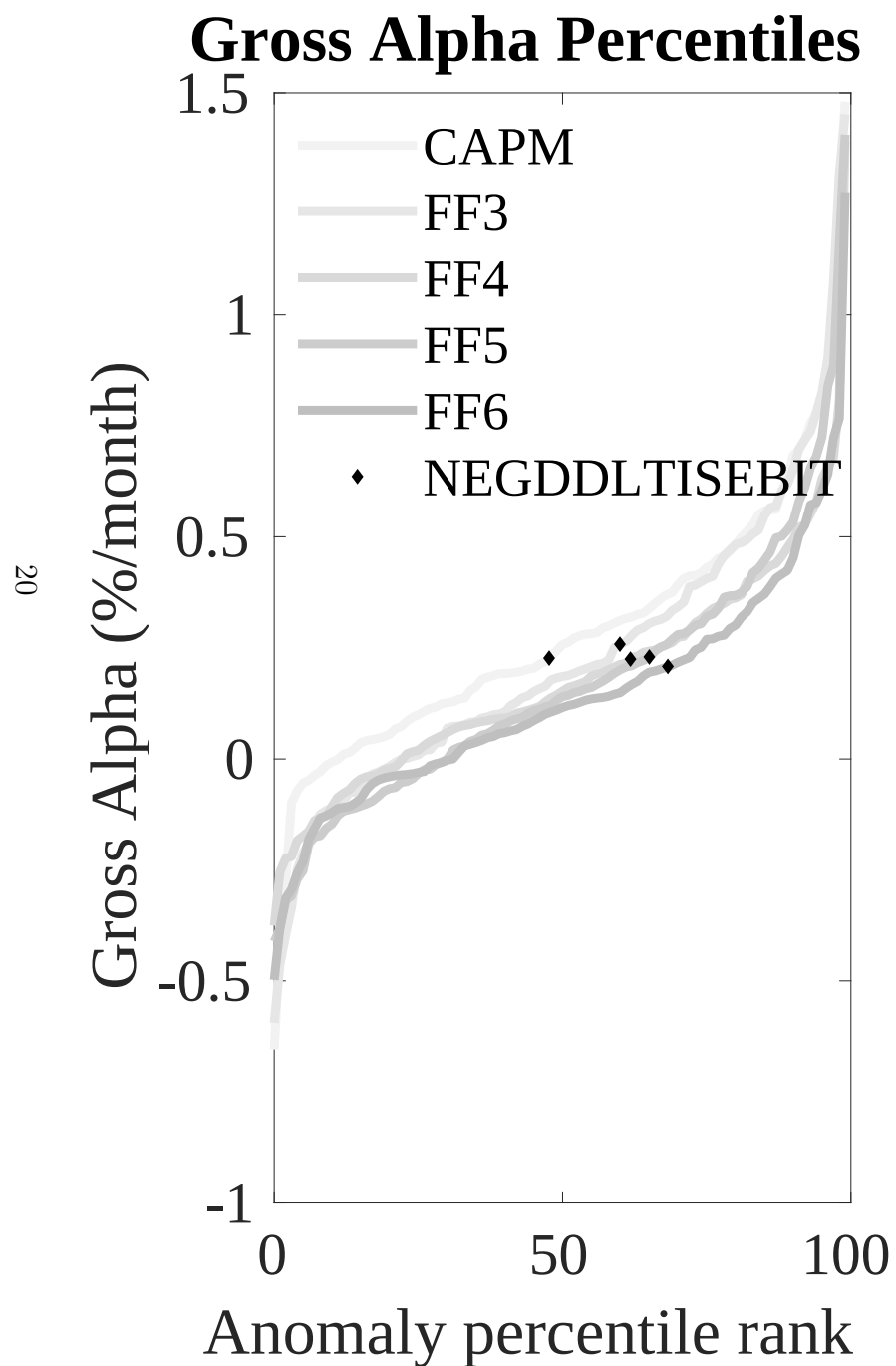


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



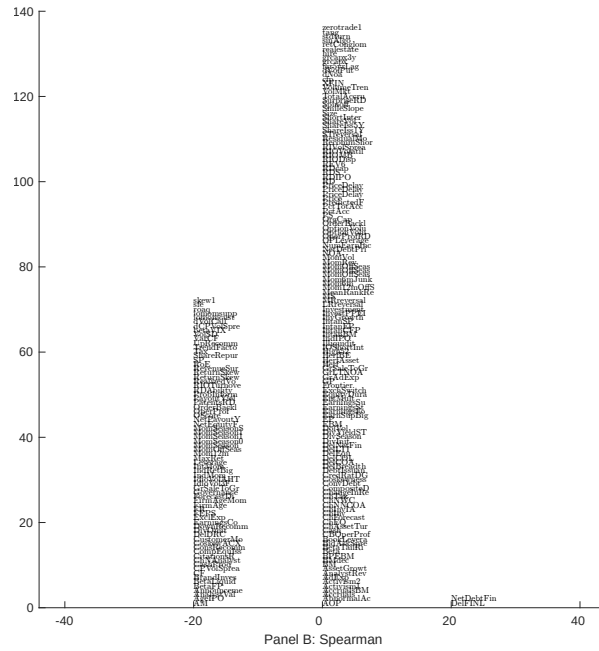
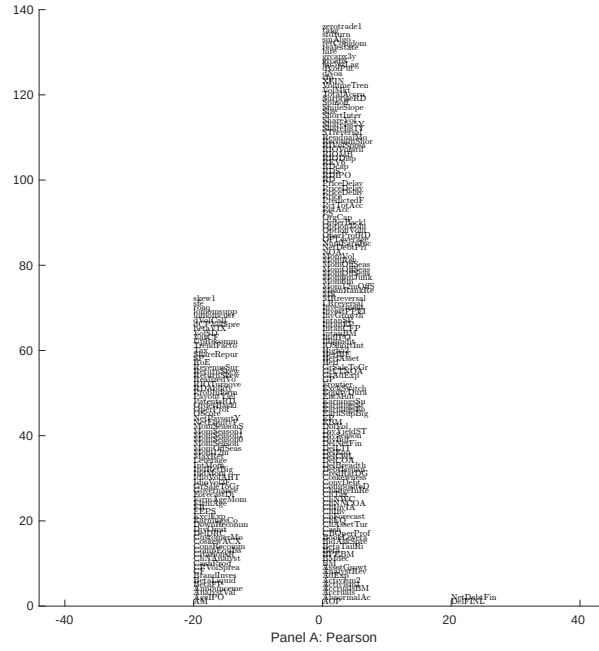


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

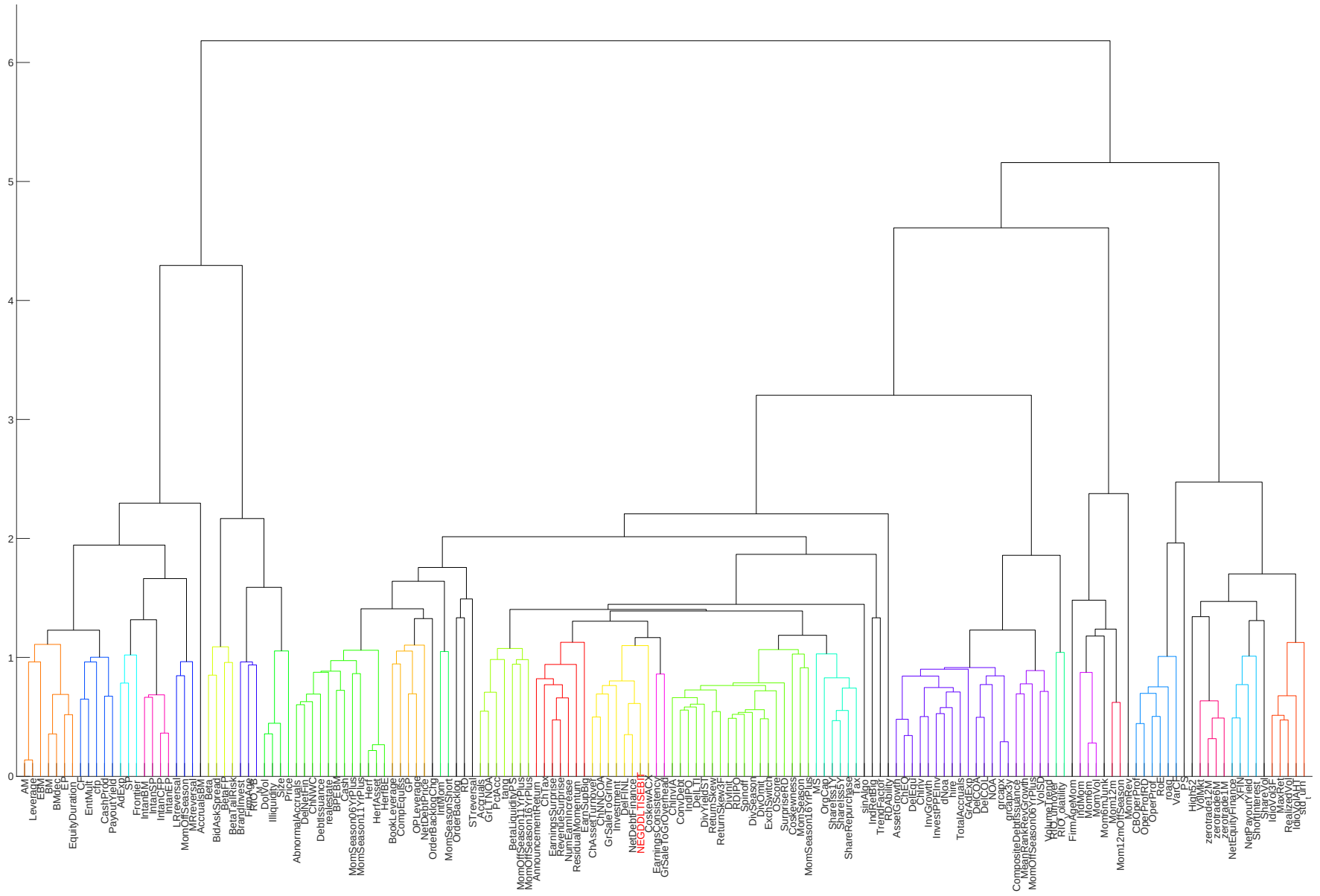


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

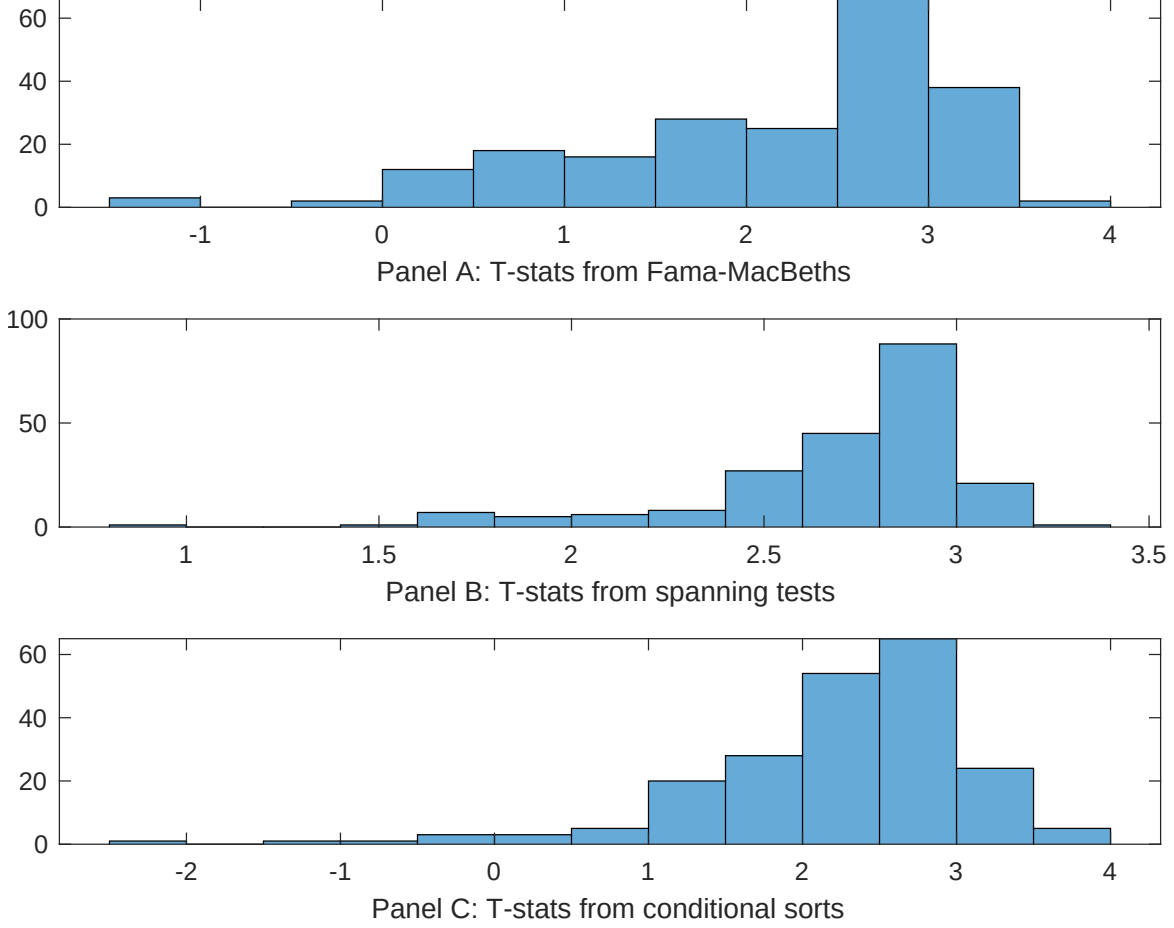


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Change in net financial assets, Investment to revenue, Accruals, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.34]	0.13 [5.30]	0.13 [5.20]	0.16 [6.29]	0.12 [4.97]	0.13 [5.16]	0.15 [6.03]
EDP	0.77 [0.20]	0.20 [0.50]	0.54 [1.38]	0.11 [2.68]	0.92 [2.34]	0.12 [2.79]	0.19 [0.43]
Anomaly 1	0.18 [9.75]						0.15 [3.08]
Anomaly 2		0.21 [9.43]					0.85 [2.03]
Anomaly 3			0.76 [5.14]				-0.85 [-2.73]
Anomaly 4				0.26 [6.05]			0.20 [4.28]
Anomaly 5					0.14 [4.49]		0.11 [2.84]
Anomaly 6						0.16 [5.16]	0.11 [3.46]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Change in net financial assets, Investment to revenue, Accruals, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.17 [2.30]	0.18 [2.40]	0.16 [2.08]	0.19 [2.46]	0.18 [2.36]	0.20 [2.57]	0.15 [1.98]
Anomaly 1	24.62 [5.72]						5.35 [0.90]
Anomaly 2		24.97 [6.02]					15.02 [2.79]
Anomaly 3			21.28 [5.92]				11.76 [2.89]
Anomaly 4				11.15 [3.72]			6.47 [2.15]
Anomaly 5					6.10 [1.98]		0.55 [0.18]
Anomaly 6						12.79 [3.86]	7.51 [2.26]
mkt	-0.59 [-0.34]	-1.11 [-0.64]	-0.76 [-0.43]	-1.40 [-0.79]	-0.70 [-0.39]	-1.44 [-0.81]	-1.01 [-0.59]
smb	-2.78 [-1.04]	-2.46 [-0.92]	0.98 [0.37]	-2.32 [-0.85]	0.91 [0.33]	-0.79 [-0.29]	-2.87 [-1.03]
hml	-13.88 [-4.11]	-14.50 [-4.31]	-15.88 [-4.72]	-14.74 [-4.30]	-13.94 [-3.95]	-15.07 [-4.41]	-13.98 [-4.12]
rmw	-7.85 [-2.28]	-8.38 [-2.43]	-3.08 [-0.90]	-4.69 [-1.35]	-3.63 [-0.99]	-6.99 [-2.00]	-6.43 [-1.78]
cma	11.42 [2.15]	13.18 [2.53]	25.09 [4.83]	18.60 [3.57]	17.67 [3.31]	18.76 [3.61]	15.72 [2.82]
umd	1.33 [0.75]	1.86 [1.06]	3.10 [1.79]	2.04 [1.13]	3.01 [1.68]	1.99 [1.10]	0.08 [0.04]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	10	10	10	7	6	7	14

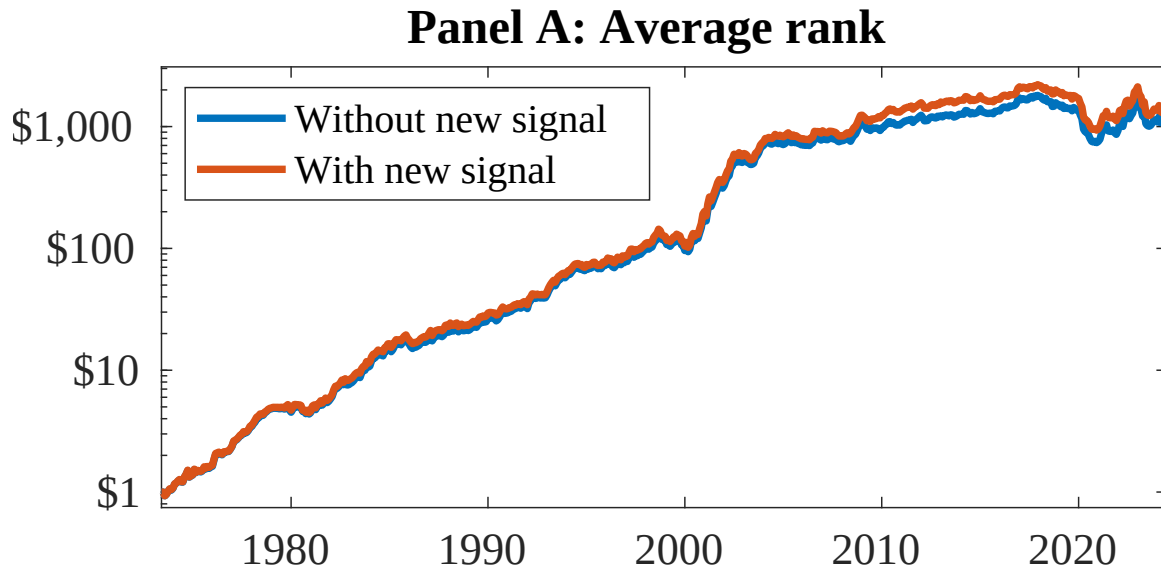


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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